

Research & Clinical Practice Literature Companion

AI in Clinical Practice — One Research Paper + One Clinical Practice Paper per Unit

This document exists to make the lectures more substantive: instead of teaching concepts in the abstract, each unit now has real, citable literature behind it — one research paper (the evidence base) and one clinical-practice-facing paper or guideline (what it means for actual care). Assign these as pre-reading, or open one on screen during the “Industry Adoption” or “Discussion” block of each unit.

A note on currency: some citations below are from 2025–2026 (verified via live search) and some are foundational/classic papers from earlier years that are still the standard reference for that concept. Verify all links before distributing to students, since URLs and availability can change.

🕒 Confirming Your Hour Structure

Yes — your math is right. Here’s the full arithmetic, explicit:

Unit	Structure	Total
Per week	3 hrs virtual classroom (theory) + 3 hrs Virtual Clinical Sandbox (lab/clinical project)	6 hrs/week
Per month (4 weeks)	3 hrs × 4 weeks (theory) + 3 hrs × 4 weeks (lab) = 12 hrs + 12 hrs	24 hrs/month
Per semester (25 weeks)	3 hrs × 25 (theory) + 3 hrs × 25 (lab) = 75 hrs + 75 hrs	150 hrs total

This matches your original blueprint exactly (12 Hours Virtual + 12 Hours Practice = 24 Hours/Month). Chapter 4 runs 5 weeks instead of 4 (to absorb the winter recess), so it totals 30 hrs instead of 24; Chapter 6 uses a different format (two 6-hour capstone shifts instead of the standard 3+3 split) — both are called out explicitly in the tables below.

📊 CHAPTER 1 — Admin Time Savings & the AI Foundation (24 hrs: 12 theory + 12 lab)

Unit 1 — Introduction to AI in Medicine

- **Research paper:** Goddard, K., Roudsari, A., & Wyatt, J.C. (2012). "Automation bias: a systematic review of frequency, effect mediators, and mitigators." *Journal of the American Medical Informatics Association*, 19(1), 121-127.
- **Clinical practice paper:** Verghese, A., Shah, N.H., & Harrington, R.A. (2018). "What This Computer Needs Is a Physician: Humanism and Artificial Intelligence." *JAMA*, 319(1), 19-20.
- **Why it matters here:** The Goddard review is the foundational evidence base for automation bias — the exact failure HITL is designed to prevent. The JAMA piece argues the clinician's judgment role doesn't disappear with AI; it changes shape.

Unit 2 — Prompt Engineering — The Four Rs

- **Research paper:** search results confirm active 2025-2026 literature on structured clinical prompting; cite via AAFP's own published framework: *American Academy of Family Physicians*, "Effective AI Prompts" (2025), *FPM* journal.
- **Clinical practice paper:** Any EHR-vendor prompt-engineering guide your institution has adopted (e.g., Epic's embedded AI documentation) — ask your IT/informatics team for the current version.
- **Why it matters here:** This is a rare case where the "clinical practice paper" is really a vendor/professional-society guideline rather than a journal article — appropriate, since prompt engineering is a skills practice, not yet a mature research literature.

Unit 3 — Ambient Scribing in Practice

- **Research paper:** Multiple 2026 studies (JAMA Network Open; 6-health-system, 263-clinician quality-improvement study) documenting burnout reduction from 51.9% to 38.8% within 30 days of ambient-scribe adoption.
- **Clinical practice paper:** A 97-encounter, 5-specialty PDQI-9 comparison study found AI-generated notes scored 4.20/5 vs. 4.25/5 for physician-authored notes — near-equivalent, with the gap concentrated in specific quality dimensions.
- **Why it matters here:** These are the exact two numbers (burnout reduction, PDQI-9 score) already built into your slide deck — now your students can see the primary literature behind them.

Unit 4 — Synthesis & Checkoff

- **Research paper:** Same as Unit 3 — re-read with focus on the "review time is non-zero" caveat.
- **Clinical practice paper:** Any of your institution's own documentation-time audit reports, if available — otherwise reuse Unit 3's sources for the checkoff discussion.
- **Why it matters here:** No new literature this week by design — the checkoff

tests synthesis of Weeks 1–3, not new content absorption.

■ CHAPTER 2 — AI-Driven Diagnostics & the Safety Override (24 hrs: 12 theory + 12 lab)

Unit 5 — Critical Appraisal of AI Diagnostics

- **Research paper:** Meta-analysis of 5 RCTs, $n=12,657$, pooled SMD = 0.182 (95% CI 0.003–0.362) for AI-CDSS diagnostic accuracy vs. standard care (2026).
- **Clinical practice paper:** News.syenza.com reporting on AI model performance vs. bedside reasoning dissociation on medical licensing-style exams.
- **Why it matters here:** This is the single most important citation in the whole course for calibrating expectations — “modest,” not “transformative,” gains.

Unit 6 — German Quality Pillars

- **Research paper:** Abbas, Q., Jeong, W., & Lee, S.W. (2025). “Explainable AI in Clinical Decision Support Systems: A Meta-Analysis of Methods, Applications, and Usability Challenges.” *Healthcare*, 13(17), 2154. DOI: 10.3390/healthcare13172154.
- **Clinical practice paper:** Salimparsa, M., et al. (2025). “Explainable AI for Clinical Decision Support Systems: Literature Review, Key Gaps, and Research Synthesis.” *Informatics*, 12(4), 119.
- **Why it matters here:** The Abbas meta-analysis (62 studies, 2018–2025) is the definitive current review of XAI methods (SHAP, Grad-CAM) across radiology, oncology, neurology, and critical care — directly grounds the Comprehensibility pillar.

Unit 7 — The Sandbox Override Exercise

- **Research paper:** 210-participant empirical study on automation bias in diagnostic decision-making — training/experience reduced false agreement; higher perceived system benefit increased it.
- **Clinical practice paper:** Active RCT protocol (ClinicalTrials.gov NCT06963957, LUMS Learning Institute) testing behavioral “nudge” interventions in physician-LLM diagnostic reasoning.
- **Why it matters here:** This is a live, currently-enrolling trial — a great discussion point for “science in progress,” not settled textbook knowledge.

Unit 8 — Synthesis & Checkoff

- **Research paper:** Re-read the 210-participant study from Unit 7, focused on the individual (not group) performance data if available in the supplementary material.
- **Clinical practice paper:** N/A — synthesis week.
- **Why it matters here:** Checkoff performance should be discussed against the same evidence base as Unit 7, since this week tests whether that evidence translated into personal behavior change.

CHAPTER 3 — Clinical Care Pathways & Data Sovereignty (24 hrs: 12 theory + 12 lab)

Unit 9 — AI-Enhanced Care Planning

- **Research paper:** Wang, Y., et al. (2018). “Clinical information extraction applications: A literature review.” *Journal of Biomedical Informatics*, 77, 34–49.
- **Clinical practice paper:** NANDA International’s published nursing diagnosis taxonomy documentation (current edition used by your institution).
- **Why it matters here:** The Wang review remains one of the most-cited overviews of clinical NLP/NER extraction methods — foundational reading before touching a real care-planning tool.

Unit 10 — GDPR & Data Sovereignty

- **Research paper:** Search-verified: Aleph Alpha (Germany) and Mistral AI (France) sovereign-hosting technical documentation; SAP + OpenAI “Delos Cloud” partnership announcement (2026) for Germany-hosted public-sector AI.
- **Clinical practice paper:** German AI Healthcare Compliance Guide 2026 (germancomplianceinstitute.com) — practitioner-facing GDPR/AI Act guidance.
- **Why it matters here:** This week’s “clinical practice paper” is necessarily a compliance guide rather than a journal article, since sovereignty is fundamentally a legal/infrastructure question.

Unit 11 — Anonymization & Secondary Data Use

- **Research paper:** Sweeney, L. (2002). “k-Anonymity: A Model for Protecting Privacy.” *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5), 557–570.
- **Clinical practice paper:** DiGA (Digital Health Applications) pathway documentation (pmc.ncbi.nlm.nih.gov/articles/PMC12973773).

- **Why it matters here:** Sweeney’s 2002 paper is THE foundational k-anonymity paper — every later anonymization framework (including the ones used in modern healthcare data pipelines) builds on it directly.

Unit 12 — Synthesis & Checkoff

- **Research paper:** Re-read Sweeney (2002) with focus on the re-identification attack examples in the original paper.
- **Clinical practice paper:** N/A — synthesis week.
- **Why it matters here:** The checkoff is essentially asking students to apply Sweeney’s 20-year-old framework to a 2026 synthetic dataset — a good discussion point on why some foundational CS theory doesn’t age out.

■ CHAPTER 4 — Ethicolegal Autonomy & the Final Checkoff (30 hrs: 15 theory + 15 lab, 5-week chapter)

Unit 13 — Liability Mapping

- **Research paper:** Price, W.N., Gerke, S., & Cohen, I.G. (2019). “Potential Liability for Physicians Using Artificial Intelligence.” *JAMA*, 322(18), 1765-1766.
- **Clinical practice paper:** Search-verified 2025/2026: Mugamba, E. (2025). “Algorithmic Malpractice: Reimagining Medical Negligence and Product Liability in AI-Assisted Clinical Decisions.” (Legal-technical model, US/Canada/EU/UK comparative review.)
- **Why it matters here:** Price/Gerke/Cohen (2019, JAMA) is the classic, most-cited starting point for AI medical liability; the 2025 Mugamba paper shows how the legal landscape has evolved since, including “dual liability exposure” (liable for using AND for not using an accurate tool).

Unit 14 — EU AI Act Risk Tiers

- **Research paper:** MDCG 2025-6 official guidance on high-risk AI classification and its relationship to existing MDR/IVDR medical device risk classes.
- **Clinical practice paper:** EU “Digital Omnibus” package (2026) — the actual deferred-deadline legislation (standalone high-risk systems: Dec 2027; embedded medical-device AI: Aug 2028).
- **Why it matters here:** This is regulatory primary source material, not academic literature — appropriate, since this week is fundamentally about knowing the actual compliance text, not someone’s interpretation of it.

Unit 15 — Informed Consent & Transparency

- **Research paper:** Kiener, M. (2020). "Artificial intelligence in medicine and the disclosure of risks." *AI & Society*, DOI: 10.1007/s00146-020-01085-w.
- **Clinical practice paper:** Patient perspectives web-based experiment: "Patients perceive AI-tool disclosure information as more important than, or similar to, standard short-term-effects disclosure." *Digital Health* (SAGE), DOI: 10.1177/20552076241247938.
- **Why it matters here:** Kiener's paper is the philosophical/ethical grounding for WHY disclosure is owed; the patient-perspectives study is empirical evidence for HOW patients actually weigh that information once given.

Unit 16-17 — Capstone Prep & Final Checkoff

- **Research paper:** Re-integrate Price/Gerke/Cohen (Unit 13), MDCG 2025-6 (Unit 14), and Kiener (Unit 15) — this week is explicitly a synthesis of all three.
- **Clinical practice paper:** Your institution's own AI-use disclosure policy, if one exists — otherwise draft one as a class exercise using the Unit 15 sources as a template.
- **Why it matters here:** The capstone dry run and checkoff should be graded against the same three-paper evidence base used across this chapter.

■ CHAPTER 5 — Consolidation & Mock OSCE (24 hrs: 12 theory + 12 lab)

Unit 18 — Integration Station Design

- **Research paper:** Anderson, L.W., & Krathwohl, D.R. (2001). *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. Longman. (The standard revision of Bloom's original 1956 taxonomy.)
- **Clinical practice paper:** Any published OSCE station-design guide from your accrediting body (most medical schools have one).
- **Why it matters here:** Anderson & Krathwohl's revised taxonomy is literally what "recall vs. application vs. analysis" (used throughout this week's slides) is drawn from.

Unit 19 — Mock OSCE, Round 1

- **Research paper:** McHugh, M.L. (2012). "Interrater reliability: the kappa statistic." *Biochemia Medica*, 22(3), 276-282.
- **Clinical practice paper:** Any of your institution's own OSCE inter-rater reliability audit reports, if available.
- **Why it matters here:** McHugh's paper is the standard, highly-cited plain-language explanation of Cohen's kappa — perfect pre-reading before this

week's scoring-reliability discussion.

Unit 20 — Mock OSCE Round 2 + Remediation

- **Research paper:** McGaghie, W.C., Issenberg, S.B., Cohen, E.R., Barsuk, J.H., & Wayne, D.B. (2011). "Does simulation-based medical education with deliberate practice yield better results than traditional clinical education? A meta-analytic comparative review." *Academic Medicine*, 86(6), 706-711.
- **Clinical practice paper:** Same source — this is both the research and practice paper for mastery-learning design, since it directly informed simulation curricula worldwide.
- **Why it matters here:** McGaghie et al. is the most-cited paper establishing that mastery-learning (fixed threshold, deliberate practice) outperforms time-based, curve-graded traditional education — the exact model Week 20's remediation policy uses.

Unit 21 — Consolidation Wrap-Up

- **Research paper:** Re-read McGaghie et al. (Unit 20) with focus on the "readiness gate before high-stakes assessment" design principle.
- **Clinical practice paper:** N/A — open forum week.
- **Why it matters here:** This week operationalizes the mastery-learning principle: catching a gap here is explicitly cheaper than catching it during the graded Capstone.

■ CHAPTER 6 — Solo Pilot Capstone & Certification (Different format: two 6-hr shifts + remediation + 1 standard week)

Unit 22-23 — Capstone Shifts 1 & 2

- **Research paper:** McGaghie et al. (2011) — same mastery-learning citation, now applied to the full-shift consistency metric.
- **Clinical practice paper:** Any published full-shift OSCE / "translational simulation" study from your specialty society, if available.
- **Why it matters here:** These two weeks are graded lab performance, not new classroom theory — see the Time/Training/Practice deck for the full 6-hour shift structure. The literature basis is the same mastery-learning framework from Chapter 5.

Unit 24 — Remediation / Retake Window

- **Research paper:** Re-read McGaghie et al. — focus on the “targeted deliberate practice” section specifically.
- **Clinical practice paper:** N/A — remediation week, scope limited to the individual student’s named gap.
- **Why it matters here:** Same principle as Unit 20 — precision remediation is evidence-based, not just administratively convenient.

Unit 25 — Certification & Course Close

- **Research paper:** Goddard, Roudsari & Wyatt (2012) — re-read as a full-course bookend, now that students have lived the automation-bias/HITL cycle for 25 weeks.
- **Clinical practice paper:** Verghese, Shah & Harrington (2018) — same bookend logic as the research paper.
- **Why it matters here:** Closing the course with the same two papers that opened it (Unit 1) reinforces that everything in between was elaboration of one idea, not 25 unrelated topics.

📖 Quick-Reference Citation List (alphabetical)

1. Abbas, Q., Jeong, W., & Lee, S.W. (2025). Explainable AI in Clinical Decision Support Systems: A Meta-Analysis. *Healthcare*, 13(17), 2154.
2. Anderson, L.W., & Krathwohl, D.R. (2001). *A Taxonomy for Learning, Teaching, and Assessing*. Longman.
3. Goddard, K., Roudsari, A., & Wyatt, J.C. (2012). Automation bias: a systematic review. *JAMIA*, 19(1), 121-127.
4. Kiener, M. (2020). Artificial intelligence in medicine and the disclosure of risks. *AI & Society*.
5. McGaghie, W.C., et al. (2011). Simulation-based medical education with deliberate practice. *Academic Medicine*, 86(6), 706-711.
6. McHugh, M.L. (2012). Interrater reliability: the kappa statistic. *Biochemia Medica*, 22(3), 276-282.
7. Mugamba, E. (2025). Algorithmic Malpractice: Reimagining Medical Negligence and Product Liability in AI-Assisted Clinical Decisions.
8. Price, W.N., Gerke, S., & Cohen, I.G. (2019). Potential Liability for Physicians Using Artificial Intelligence. *JAMA*, 322(18), 1765-1766.
9. Salimparsa, M., et al. (2025). Explainable AI for Clinical Decision Support Systems. *Informatics*, 12(4), 119.
10. Sweeney, L. (2002). k-Anonymity: A Model for Protecting Privacy. *Int. J. Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5), 557-570.
11. Verghese, A., Shah, N.H., & Harrington, R.A. (2018). What This Computer Needs Is a Physician. *JAMA*, 319(1), 19-20.
12. Wang, Y., et al. (2018). Clinical information extraction applications: A literature review. *J. Biomedical Informatics*, 77, 34-49.

Plus regulatory primary sources: MDCG 2025-6 guidance, EU AI Act “Digital

Omnibus” package (2026), and the German AI Healthcare Compliance Guide 2026 — these are official documents, not peer-reviewed papers, and should be sourced directly from the issuing body.

Instructor Note

Two citations recur most often (Goddard 2012 on automation bias, and the HITL principle they support) — this is intentional. A course that cites 25 unrelated papers teaches 25 facts; a course that keeps returning to the same 2-3 foundational papers in new contexts teaches one transferable framework. Use that repetition explicitly in class: “we saw this same automation-bias paper in Week 1 — now watch it show up again in Week 7’s override exercise.”